

## ANCHORAGE LAKE HOOD, AK, USA

WMO: 702725

Lat: 61.178N

Lon: 149.966W

Elev: 90

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26491

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.6       | -2.6 | -15.8                         | 2.3 | -3.8 | -11.5 | 3.0 | 0.1  | 20.5                  | 23.0 | 18.2 | 23.8 | 1.6                   | 340  | 0.637 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.1                   | 74.7            | 60.1 | 71.4 | 58.5 | 68.3 | 57.2 | 61.6                | 72.2 | 59.9 | 68.8 | 58.5 | 66.1 | 6.2                  | 310  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 56.9                            | 69.3 | 64.2 | 55.4 | 65.6 | 62.8 | 54.4 | 63.2 | 61.9 | 27.4          | 72.2 | 26.3 | 69.2 | 25.3 | 66.1 | 69.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 18.3          | 15.8 | 12.9 | -10.0                      | 80.1 | 6.0                | 4.2 | -14.3                                              | 83.2 | -17.8      | 85.6 | -21.1      | 88.0 | -25.5      | 91.1 |    |    |
| (5)               |      |      | -10.7                      | 64.1 | 5.8                | 2.4 | -14.8                                              | 65.9 | -18.2      | 67.3 | -21.4      | 68.6 | -25.6      | 70.4 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual                           | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|----------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)                              | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                            | 38.7   | 17.9  | 21.7  | 26.1  | 38.3  | 49.6  | 57.8  | 61.2  | 58.7  | 50.5  | 37.9  | 24.0  | 19.6  | (6)  |
| (7)  |                                                                                  | DBStd                            | 17.59  | 12.10 | 10.03 | 8.65  | 6.05  | 5.37  | 4.33  | 3.81  | 3.73  | 5.12  | 7.90  | 9.81  | 10.23 | (7)  |
| (8)  |                                                                                  | HDD50                            | 5105   | 994   | 791   | 740   | 353   | 71    | 1     | 0     | 1     | 55    | 377   | 780   | 942   | (8)  |
| (9)  |                                                                                  | HDD65                            | 9616   | 1459  | 1211  | 1205  | 802   | 478   | 220   | 131   | 199   | 434   | 840   | 1230  | 1407  | (9)  |
| (10) |                                                                                  | CDD50                            | 983    | 0     | 0     | 0     | 1     | 59    | 235   | 346   | 270   | 70    | 2     | 0     | 0     | (10) |
| (11) |                                                                                  | CDD65                            | 22     | 0     | 0     | 0     | 0     | 0     | 5     | 13    | 4     | 0     | 0     | 0     | 0     | (11) |
| (12) |                                                                                  | CDH74                            | 124    | 0     | 0     | 0     | 0     | 4     | 36    | 65    | 19    | 0     | 0     | 0     | 0     | (12) |
| (13) |                                                                                  | CDH80                            | 12     | 0     | 0     | 0     | 0     | 0     | 3     | 8     | 1     | 0     | 0     | 0     | 0     | (13) |
| (14) | Wind                                                                             | WSAvg                            | 4.9    | 4.4   | 4.3   | 5.2   | 5.0   | 5.9   | 5.8   | 5.1   | 4.6   | 4.7   | 4.9   | 4.9   | 4.5   | (14) |
| (15) | Precipitation                                                                    | PrecAvg                          | 16.4   | 0.8   | 0.8   | 0.7   | 0.5   | 0.7   | 1.1   | 1.8   | 2.8   | 3.0   | 1.9   | 1.2   | 1.1   | (15) |
| (16) |                                                                                  | PrecMax                          | 27.6   | 2.7   | 2.4   | 2.8   | 1.9   | 1.9   | 3.4   | 4.5   | 9.8   | 7.7   | 4.3   | 2.8   | 2.7   | (16) |
| (17) |                                                                                  | PrecMin                          | 8.1    | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.3   | 0.7   | 0.4   | 0.0   | 0.1   | (17) |
| (18) |                                                                                  | PrecStd                          | 3.5    | 0.6   | 0.5   | 0.5   | 0.5   | 0.4   | 0.7   | 1.0   | 1.8   | 1.7   | 0.9   | 0.8   | 0.7   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                             | DB     | 45.3  | 43.9  | 45.9  | 59.1  | 73.0  | 78.6  | 79.9  | 76.5  | 65.8  | 56.2  | 46.5  | 45.6  | (19) |
| (20) |                                                                                  |                                  | MCWB   | 38.4  | 37.1  | 38.6  | 46.4  | 54.2  | 61.3  | 63.0  | 61.8  | 54.2  | 48.1  | 40.4  | 38.6  | (20) |
| (21) |                                                                                  | 2%                               | DB     | 39.8  | 39.4  | 42.8  | 53.6  | 67.0  | 73.1  | 75.1  | 72.0  | 62.5  | 51.8  | 41.7  | 40.0  | (21) |
| (22) |                                                                                  |                                  | MCWB   | 35.1  | 35.1  | 36.3  | 43.1  | 51.4  | 58.3  | 61.3  | 59.7  | 53.0  | 45.2  | 37.9  | 35.8  | (22) |
| (23) |                                                                                  | 5%                               | DB     | 36.1  | 36.5  | 39.9  | 50.3  | 62.8  | 69.8  | 71.7  | 68.4  | 60.3  | 49.7  | 38.9  | 35.4  | (23) |
| (24) |                                                                                  |                                  | MCWB   | 33.3  | 33.6  | 34.5  | 40.9  | 49.5  | 56.8  | 59.8  | 58.1  | 51.8  | 44.2  | 36.2  | 32.9  | (24) |
| (25) |                                                                                  | 10%                              | DB     | 32.6  | 34.0  | 37.4  | 47.9  | 59.1  | 66.2  | 68.6  | 65.7  | 57.8  | 47.5  | 36.3  | 32.7  | (25) |
| (26) |                                                                                  |                                  | MCWB   | 30.5  | 31.7  | 32.8  | 39.7  | 47.9  | 55.0  | 58.4  | 56.5  | 50.9  | 43.2  | 33.9  | 30.9  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%                             | WB     | 39.1  | 37.6  | 39.3  | 47.5  | 55.2  | 62.5  | 64.4  | 63.3  | 55.7  | 49.9  | 41.1  | 38.8  | (27) |
| (28) |                                                                                  |                                  | MCD B  | 44.8  | 42.5  | 44.7  | 57.9  | 71.3  | 75.7  | 77.1  | 73.7  | 62.2  | 54.3  | 44.5  | 44.2  | (28) |
| (29) |                                                                                  | 2%                               | WB     | 35.5  | 35.5  | 37.0  | 43.8  | 52.2  | 59.5  | 62.1  | 60.8  | 54.3  | 46.5  | 38.7  | 36.0  | (29) |
| (30) |                                                                                  |                                  | MCD B  | 39.1  | 38.5  | 41.5  | 52.2  | 64.6  | 70.5  | 73.4  | 69.6  | 59.9  | 50.1  | 41.6  | 39.6  | (30) |
| (31) |                                                                                  | 5%                               | WB     | 33.2  | 33.8  | 35.2  | 41.7  | 50.4  | 57.5  | 60.5  | 59.2  | 53.2  | 45.2  | 36.3  | 33.1  | (31) |
| (32) |                                                                                  |                                  | MCD B  | 35.9  | 36.3  | 39.3  | 49.0  | 60.8  | 67.1  | 69.9  | 66.7  | 58.1  | 48.6  | 38.6  | 35.1  | (32) |
| (33) |                                                                                  | 10%                              | WB     | 30.6  | 31.8  | 33.3  | 40.1  | 48.8  | 56.0  | 59.1  | 57.7  | 52.0  | 43.6  | 34.0  | 30.9  | (33) |
| (34) |                                                                                  |                                  | MCD B  | 32.6  | 34.1  | 36.9  | 47.0  | 58.1  | 64.9  | 67.0  | 63.9  | 56.6  | 47.0  | 36.0  | 32.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB                            | MDBR   | 10.4  | 11.1  | 13.8  | 14.5  | 14.5  | 13.9  | 12.1  | 12.0  | 11.6  | 9.6   | 9.3   | 10.0  | (35) |
| (36) |                                                                                  |                                  | MCDBR  | 13.0  | 11.4  | 13.7  | 18.1  | 20.6  | 19.9  | 18.1  | 17.2  | 14.7  | 10.9  | 10.0  | 12.3  | (36) |
| (37) |                                                                                  |                                  | MCWBR  | 9.5   | 8.4   | 9.0   | 10.0  | 9.9   | 9.2   | 8.3   | 8.0   | 7.7   | 6.4   | 7.2   | 9.3   | (37) |
| (38) |                                                                                  | 5% WB                            | MCD BR | 12.4  | 10.7  | 12.7  | 16.7  | 18.4  | 17.5  | 16.2  | 15.0  | 12.2  | 9.3   | 9.6   | 11.5  | (38) |
| (39) |                                                                                  |                                  | MCWBR  | 9.3   | 8.0   | 8.5   | 9.5   | 9.3   | 8.5   | 7.8   | 7.4   | 6.9   | 5.8   | 7.1   | 9.1   | (39) |
| (40) |                                                                                  | Clear-Sky<br>Solar<br>Irradiance | taub   | 0.234 | 0.287 | 0.306 | 0.347 | 0.369 | 0.358 | 0.372 | 0.356 | 0.318 | 0.298 | 0.243 | N/A   | (40) |
| (41) | taud                                                                             |                                  | 2.131  | 2.234 | 2.272 | 2.258 | 2.318 | 2.430 | 2.418 | 2.468 | 2.563 | 2.481 | 2.133 | N/A   | (41)  |      |
| (42) | Ebn at Noon                                                                      |                                  | 161    | 220   | 258   | 267   | 270   | 276   | 268   | 264   | 257   | 221   | 150   | N/A   | (42)  |      |
| (43) | Edh at Noon                                                                      |                                  | 12     | 21    | 28    | 35    | 36    | 32    | 32    | 28    | 22    | 16    | 11    | N/A   | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg                           | 91     | 308   | 822   | 1357  | 1636  | 1689  | 1498  | 1185  | 772   | 360   | 132   | 47    | (44)  |      |
| (45) |                                                                                  | RadStd                           | 13     | 45    | 90    | 121   | 151   | 110   | 134   | 131   | 108   | 43    | 21    | 6     | (45)  |      |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (26 neighbors) | N/A     | +3.60  | +3.06   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page